Ahmad M. Alkadri

Researcher | Engineer | Mathematician

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Google Scholar | LinkedIn

Postdoctoral scholar working in continuum mechanics, scientific computing, and quantum algorithms. Develops mathematical models, numerical methods, scientific software, and verification-oriented engineering workflows, with an emphasis on making assumptions and evidence explicit.

Appointments and experience

Postdoctoral Scholar — **Sep. 2025 – present**
University of California, Berkeley — *Berkeley, CA*

- Develop quantum and classical algorithms for discretized partial differential equations and high-dimensional dynamics, including finite-element assembly, boundary conditions, and scientific observables.
- Develop tensor-network and Monte Carlo methods for high-dimensional expectations and integrals arising in PDE and stochastic models.
- Build scientific software and agentic-engineering workflows that integrate specifications, tests, provenance, documentation, and bounded recovery paths.

Graduate Student Researcher — **Aug. 2019 – Aug. 2025**
University of California, Berkeley — *Berkeley, CA*

- Derived continuum theories for curved, deforming lipid membranes, including mechanics, permeability, osmosis, and multicomponent bulk-fluid coupling.
- Developed numerical methods for nonlinear PDEs on evolving surfaces using finite-element methods, Python, C++, FEniCS, and arbitrary Lagrangian–Eulerian formulations.
- Analysed nonlinear dynamical systems using perturbation theory, eigenvalue methods, and asymptotic reasoning.

Chemical Engineering Research Intern — **May 2017 – Aug. 2018**
NOVA Chemicals — *Calgary, Canada*

- Developed process models and data-analysis tools in Python, MATLAB, C++, Aspen Plus, and Excel to support polymer-plant troubleshooting and optimization.
- Built a temperature-anomaly detector and collaborated on phase-equilibrium and heat- and mass-transfer modelling.

Research Intern — **May 2016 – Sep. 2016**
Pharmaceutical Production Research Facility — *Calgary, Canada*

- Supported experiments on mechanical and chemical signals in stem-cell differentiation and developed experience in bioprocessing, statistical analysis, and scientific literature review.

Education

Ph.D., Chemical Engineering, University of California, Berkeley — 2025
M.A., Mathematics, University of California, Berkeley — 2025
B.Sc., Pure and Applied Mathematics, University of Calgary — 2019
B.Sc., Chemical Engineering, University of Calgary — 2019

Technical expertise

Scientific computing: finite-element methods, numerical PDEs, numerical linear algebra, optimization, tensor-network and Monte Carlo methods.
Physical modelling: continuum mechanics, surface mechanics, thermodynamics, transport phenomena, stability analysis.
Programming and tools: Python, C++, Mathematica, MATLAB, FEniCS, Git, LaTeX; testing, reproducible builds, and agentic-engineering workflows.
Languages: English (native), Arabic (fluent), French (basic).

Selected projects

Qu-FEM *Public research code*
Mathematica notebooks accompanying A Quantum Algorithm for the Finite Element Method; reproduces numerical demonstrations and explicit circuits for finite-element assembly and PDE examples. Repository.

MathAnimate *Private alpha prototype*
An agentic compiler for explanatory Manim videos with structured storyboard planning, static and rendering checks, bounded repair, protected capture and replay, and reviewable provenance bundles.

Chemical Thermodynamics *Early-stage public library*
A Python package with an SI-unit API for component data, equations of state, and temperature–pressure flash calculations, supported by examples and tests. Repository.

Agentic Engineering: An Introduction *Public workshop*
A demo-driven workshop on engineered tasks, acceptance tests, prompts, recovery loops, verification, and human ownership of technical decisions. Repository.

Publications

Journal articles

- **A. M. Alkadri** and K. K. Mandadapu. "Irreversible thermodynamics of curved lipid membranes. II. Permeability and osmosis." *Physical Review E* 112, 034413 (2025). DOI.
- T. Kharazi, **A. M. Alkadri**, J.-P. Liu, K. K. Mandadapu, and K. B. Whaley. "Explicit block encodings of boundary value problems for many-body elliptic operators." *Quantum* 9, 1764 (2025). DOI.

Preprints

- T. Kharazi, **A. M. Alkadri**, K. K. Mandadapu, and K. B. Whaley. "Provable Quantum Speedups for Reaction-Rate Estimation in High-Dimensional Fokker–Planck Dynamics." arXiv:2601.15523v2 (2026). arXiv.
- **A. M. Alkadri**, T. D. Kharazi, K. B. Whaley, and K. K. Mandadapu. "A Quantum Algorithm for the Finite Element Method." arXiv:2510.18150v1 (2025). arXiv.

Selected presentations

- "A Quantum Algorithmic Framework for the Finite Element Method." Oral presentation, 17th World Congress on Computational Mechanics and 10th European Congress on Computational Methods in Applied Sciences and Engineering (WCCM-ECCOMAS 2026), Munich, Germany, July 2026.
- "A Quantum Algorithm for the Finite Element Method (Qu-FEM)." Oral presentation, American Physical Society March Meeting, Anaheim, CA, March 2025.
- "Theory and Stability of Permeable Membrane Tubes Undergoing Osmotic Shocks." Oral presentation, UC Berkeley Chemical and Biomolecular Engineering Research Symposium, April 2023.
- "Stability Analysis of Permeable Membrane Tubes Undergoing Osmotic Shocks." Oral presentation, AIChE Annual Meeting, Phoenix, AZ, November 2022.

Selected honours and service

NSERC Postgraduate Scholarship–Doctoral	2020
Chemical Engineering Graduate Fellowship, UC Berkeley	2019
Intern of Merit, Schulich School of Engineering	2018
25th Anniversary Scholarships in Mathematics and Statistics, University of Calgary	2017

Service and outreach: KALX 90.7 FM Berkeley volunteer (2026–present); Bay Area Scientists Inspiring Students volunteer (2020–2023); WCCUSD STEM Fair volunteer judge (2022); AIChE University of Calgary student chapter vice president (2018–2019).